Physics - Graduate Level Assessment 10 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Polarization is a property of

   (a) mechanical waves
   (b) sound waves
   (c) all waves
   (d) None of these
   (e) radio waves

2. Which of these does NOT belong in the family of electromagnetic waves?

   (a) Sound
   (b) Ultraviolet waves
   (c) Infrared waves
   (d) Microwaves
   (e) X-rays

3. The magnitude of the gravitational force on a satellite is constant if the orbit is

   (a) oval
   (b) spiral
   (c) elliptical
   (d) All of these
   (e) rectangular

4. Two thin convex lenses of focal lengths f_1 and f_2 are separated by a distance equal to 2 f_2. If f_1 = 3 f_2, what is the focal length of the combination?

(3 f_2) / 2

   (a) (f_1 + f_2) / 2
   (b) (2 f_2) / 3
   (c) 2 f_2
   (d) (3 f_2) / 3

5. According to the BCS theory, the attraction between Cooper pairs in a superconductor is due to

   (a) Magnetic attraction
   (b) vacuum polarization

(c) interactions with the ionic lattice

(d) Gravitational pull

(e) Electrostatic attraction

True / False (10 marks)

Answer True or False

6. True or False: Relativistic equations for time dilation and length contraction only apply at speeds greater than the speed of light.
7. True or False: The current through the filament of a light bulb with a resistance of 240 ohms at 120 volts is 2 amperes.
8. True or False: The total spin quantum number of the electrons in the ground state of neutral nitrogen (Z = 7) is 3.
9. True or False: Jupiter's internal heat is largely produced by nuclear fusion occurring in its core, similar to that of stars.
10. True or False: The buoyant force on the balloon decreases as it sinks deeper into the lake due to increased water pressure.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the thermal expansion considerations in the design of a two-mile-long steel bridge subjected to temperature variations from -$40^t o+110' calculating the total expansion and its implic$

11. Calculate the total charge in coulombs for a mass of 75.0 kg of electrons, and discuss the implications of this charge in the context of electrostatics and real-world applications.

End of Paper